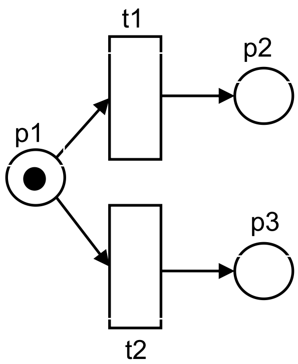
Question 1: Petri Nets 1

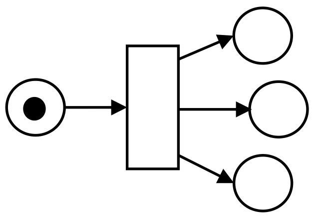
Answer the following questions:

a) Draw a parallel flow from p1 to p4. P1 should initially contain two tokens, and p4 should finally contain two tokens. A Maximum of 4 places are allowed.
(2 minutes)

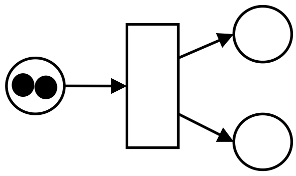
b) Add a final state to the petri net by adding whatever you want, so that an equivalent BPMN would be valid and block-structured. Write the equivalent BPMN down with the correct labels.
(2 minutes)



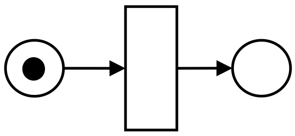
c) Make the net 3-bounded and live. Add only transitions and flows. Capacities are not allowed, edge weights only on newly added flows.
(5 minutes)



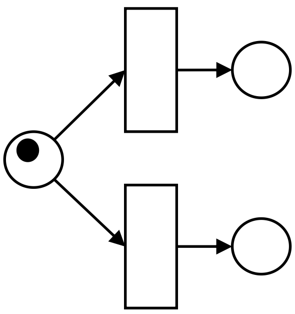
d) Make the net live so that ALWAYS two tokens are in the net. Do not add any new places.
(1 minute)



e) Introduce a deadlock into the net. For every concept (P,T,F,C,W) you add, 1 point is deducted.
(2 minutes, max 2 points)



f) Make the net not sound.
(1 minute)



g) Draw a minimal petri net with a lifelock.

h) Define structural soundness and give an example for a minmal structural sound petri net.

i) How is an EC net different from a PT net? What extensions does a PT net have?

Question 2: Petri Nets 2

Given the following Petri Net:

$N = (P,T,F,C,W,M_0)$
P = {p1, p2, p3, p4, p5}
T = {t1, t2, t3, t4}
F = {(p1, t1), (t1, p2), (t1, p3), (t1, p4), (t1, p5), (p3, t3), (p4, t2), (t2, p5), (t3, p5),
(p2, t4), (p5, t4), (t4, p1)}
C = {}
W = {(p5, t4, 3)}
M_0 = {(p1, 1)}

a) Draw the Petri Net! (1 minutes)
b) Perform a Reachability Analysis! (7 minutes)

For each of the following questions, always refer to the reachability analysis. All answers that to not refer to values in the reachability analysis will be not awarded with points (i.e. definitions result in 0 points).

c) What kind of Petri Net is this? Why? (1 minute)

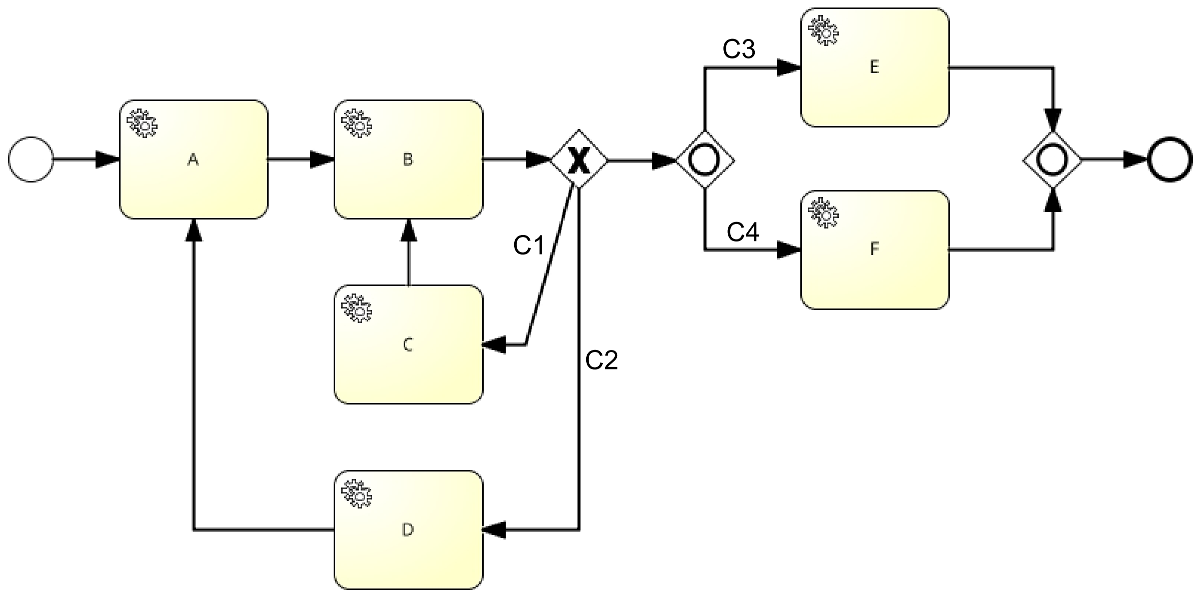
d) Is the petri net live? Why? (1 minute)

e) Is the petri net safe? Why? (1 minute)

f) Is the petri net k-bounded? Why? What is k? (1 minute)

Question 3: Model Transformation (12 Minutes)

Given the following model, write down a pseudo code program, that describes the model. **The result should be free of race conditions and potential deadlocks.** Please use visible indentation for your pseudo code.



Use the following (and only the following) pseudo code statements:

Decision	if CONDITION else if CONDITION else end
Loop	while CONDITION && CONDITION end
Task	task LABEL

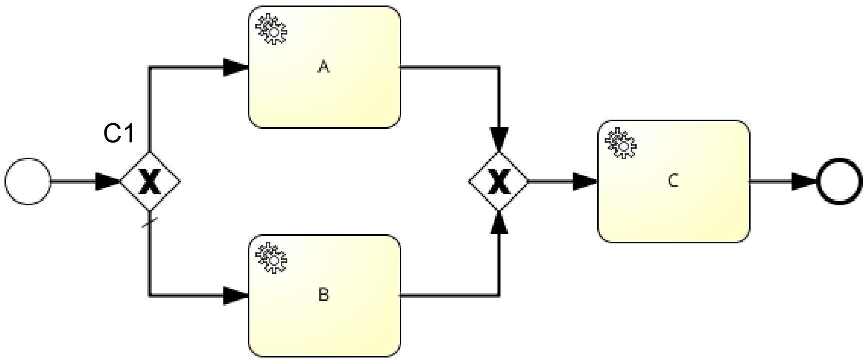
Parallel	parallel
	  branch
	    …
	  end
	  branch
	    …
	  end
	end

Break out of innermost loop	break
Set variables to true	set CONDITION_NAME

If you use “set CX1” then CX1 can be used in an if statement to evaluate to true (i.e. “if CX1 …. end”).

Everything else you might see (inclusive gateway, event gateway), has to be expressed in terms of the above statements.

Example:



if C1

  task A

else

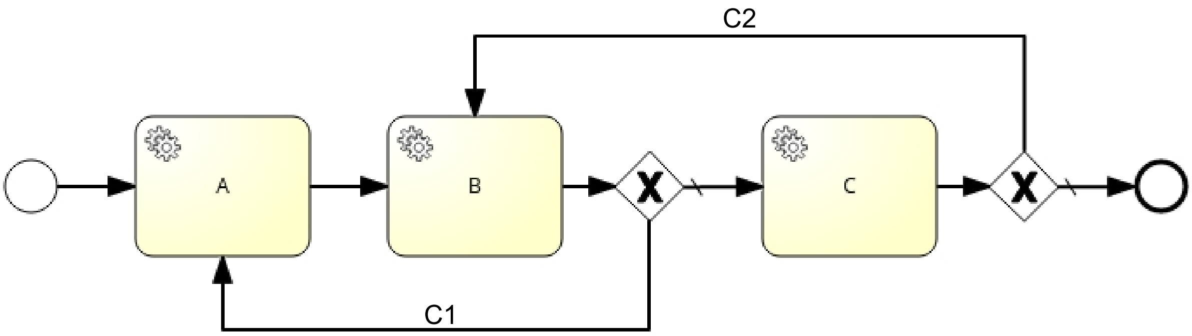
  task B

end

task C

Question 4: Model Transformation (12 Minutes)

Given the following model, write down a pseudo code program, that describes the model. **The result should be free of race conditions and potential deadlocks.** Please use visible indentation for your pseudo code.



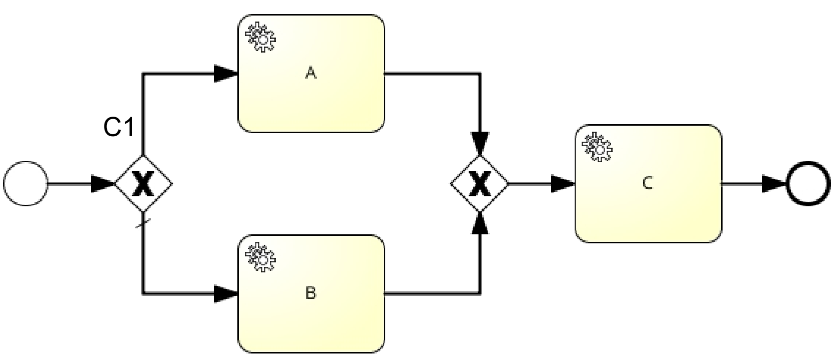
Use the following (and only the following) pseudo code statements:

Decision	if CONDITION else if CONDITION else end
Loop	while CONDITION && CONDITION end
Task	LABEL

Parallel	parallel
	  branch
	    …
	  end
	  branch
	    …
	  end
	end

Break out of innermost loop	break
Set variables to boolean value	set VAR true; set VAR false

Valid conditions in the example above are: C1, C2. Additionally TRUE and FALSE are valid conditions. Everything you see in the diagram has to be expressed in terms of the above statements. Example:




```
if C1
  A
else
  B
end
C
```

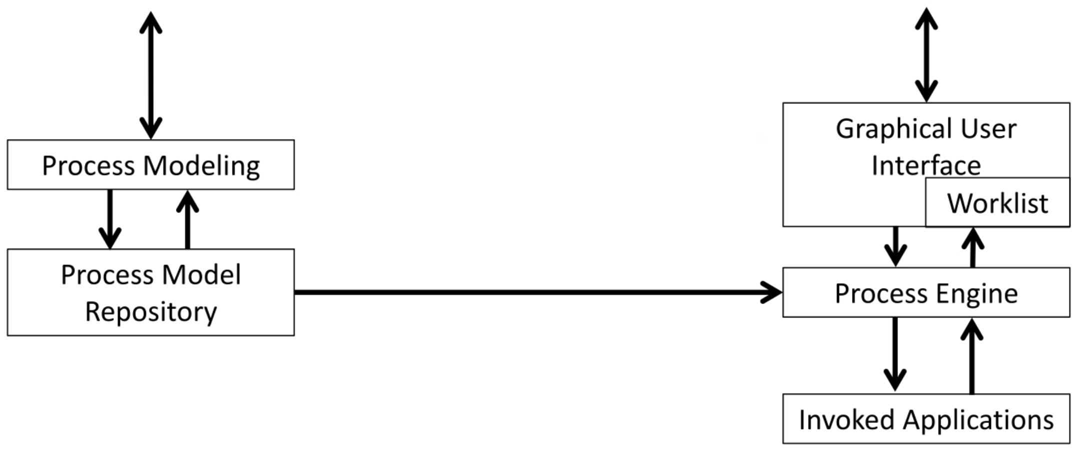
Question 5: Petrinets 1 (20 Minutes)

A person sits in a small beer selling kiosk in front of the University. The person has one bottle of beer. The person wants to sell this bottle to a customer. Sometimes a customer enters the kiosk and inspects the beer bottle. A customer that likes the beer pays and leaves. A customer that does not like the beer leaves The seller just sits there and takes the money, if the beer is sold.

a) Draw an E/C net. The seller and the customer should be represented by their own token in the net. Each transition is to be named tx: yyyy with yyyy being a label and x being a number.
b) Make the net live and describe how the scenario changes.

Question 6: Free Answers (3 minutes each)

a) How do humans interact with processes? How can work be distributed? What is the minimal input that this mechanism needs?
b) What is the purpose of a correlator?
c) What is the difference between a process choreography and a process orchestration?
d) Which interaction patterns between a task and its implementation exist?
e) Describe a typical lifecycle of process instance.
f) Describe a typical lifecycle of process task.
g) Describe a typical lifecycle of worklist.
h) What is the minimal information an item in a process log has to carry?
i) What is the difference between logging and monitoring?
j) What are potential retention policies, when a correlator has to store a message?
k)



What are the stakeholders in a process management system?

l) Why are process logs important? (answer with one sentence) For what can they be used (answer with one less than 3 words).

Question 7: Log (3 minutes)

```
<trace>
  <string key="concept:name" value="(1)"/>
  <string key="creator" value="Juergen Mangler"/>
  <event>
    <string key="concept:name" value="(2)"/>
    <string key="(3)" value="(4)"/>
    <string key="(5)" value="(6)"/>
    <date key="(7)" value="(8)"/>
  </event>
  ...
</trace>
```

Name the attributes 3,5,7. What is there purpose? Why are these and concept name on the trace level important? What are possible values for 1,2,4,6,8?